

Name: A. Abhishek Reddy

Rg no:192424192

DATE:29/7/2026

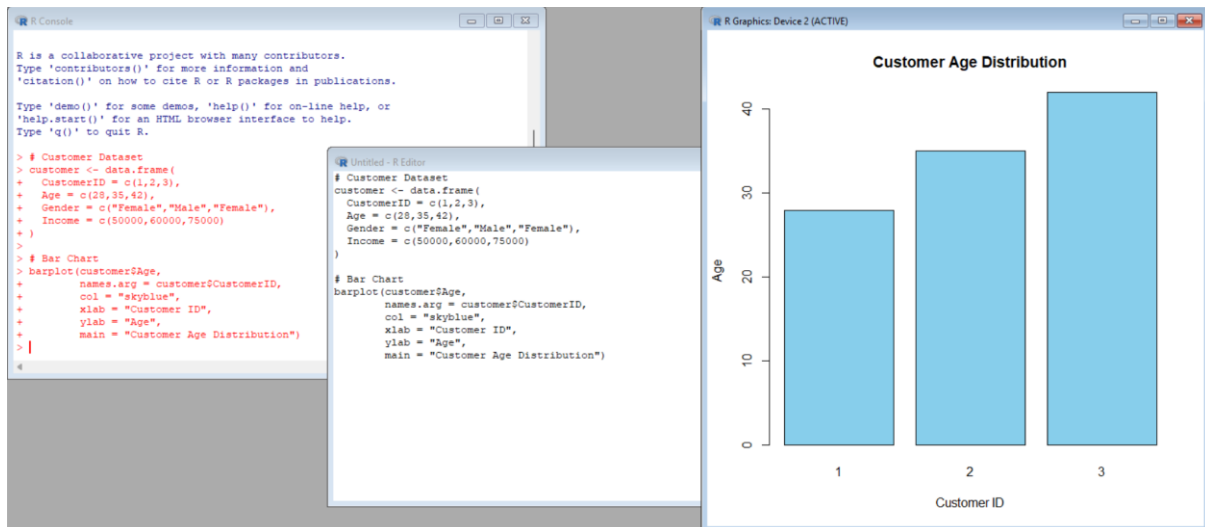
Set 16: Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

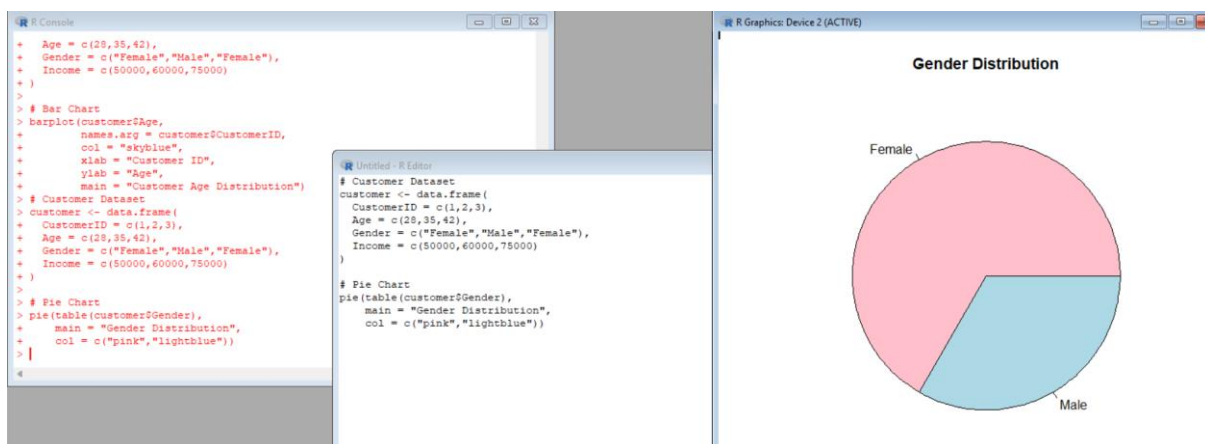
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
customer <- data.frame(  
  CustomerID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)  
  
barplot(customer$Age,  
  names.arg = customer$CustomerID,  
  col = "skyblue",  
  xlab = "Customer ID",  
  ylab = "Age",  
  main = "Customer Age Distribution")
```



2. Generate a pie chart to represent the distribution of customers by gender.

```
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
pie(table(customer$Gender),
  main = "Gender Distribution",
  col = c("pink","lightblue"))
```



3. Build a table to show the demographic information for each customer. Label the table elements.

```
customer <- data.frame(
  CustomerID = c(1,2,3),
```

```

Age = c(28,35,42),
Gender = c("Female","Male","Female"),
Income = c(50000,60000,75000)
)
print(customer)

```

```

  CustomerID Age Gender Income
1          1  28 Female  50000
2          2  35  Male  60000
3          3  42 Female  75000

```

4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics

```

library(shiny)

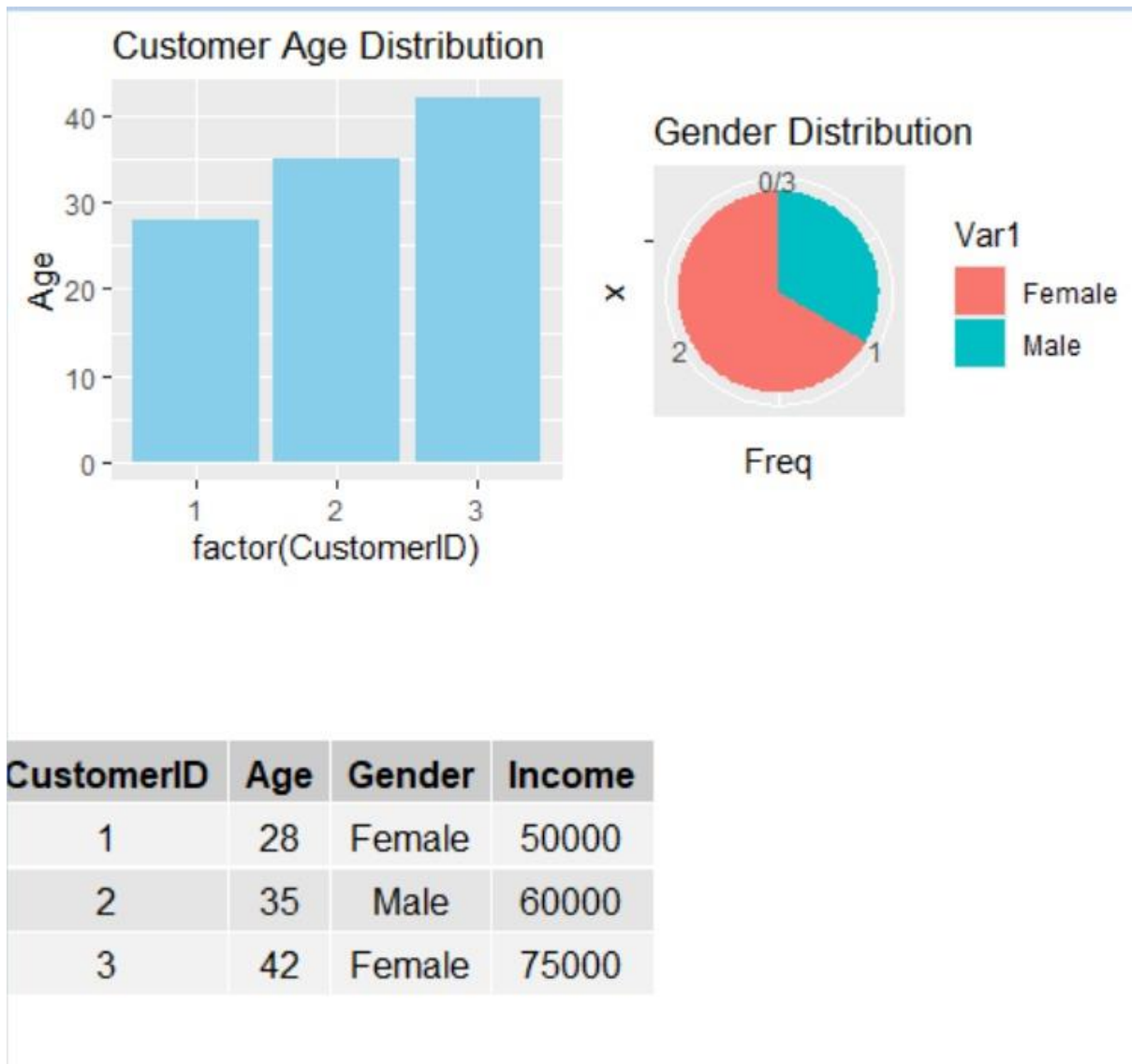
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

ui <- fluidPage(
  titlePanel("Customer Demographics Dashboard"),
  tableOutput("table")
)

server <- function(input, output) {
  output$Table <- renderTable(customer)
}

shinyApp(ui = ui, server = server)

```



Set 17: Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

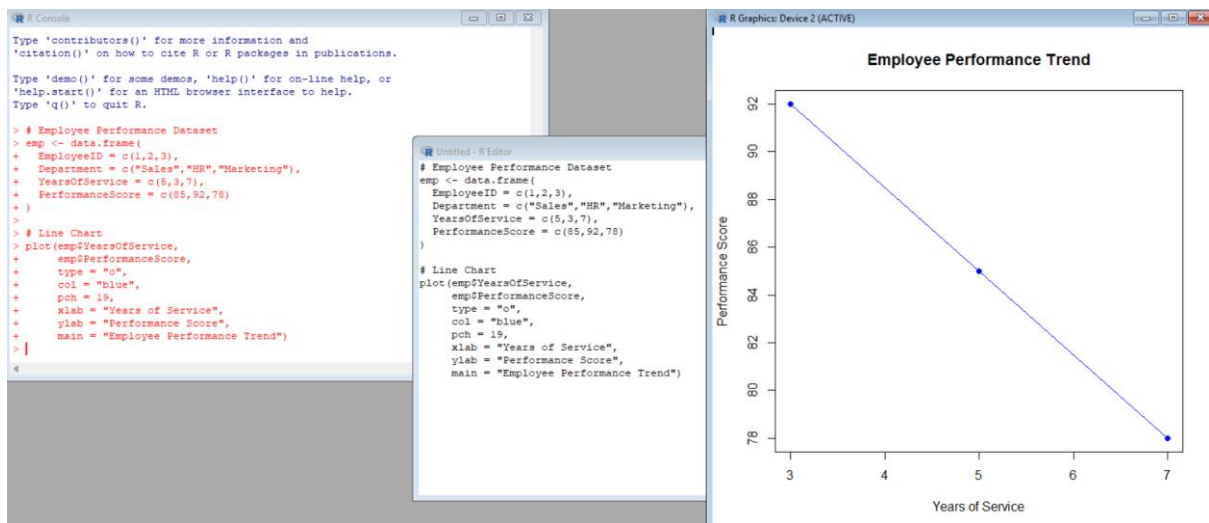
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
emp <- data.frame(
```

```

EmployeeID = c(1,2,3),
Department = c("Sales","HR","Marketing"),
YearsOfService = c(5,3,7),
PerformanceScore = c(85,92,78)
)
plot(emp$YearsOfService,
     emp$PerformanceScore,
     type = "o",
     col = "blue",
     pch = 19,
     xlab = "Years of Service",
     ylab = "Performance Score",
     main = "Employee Performance Trend")

```



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```

emp <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
barplot(table(emp$Department),

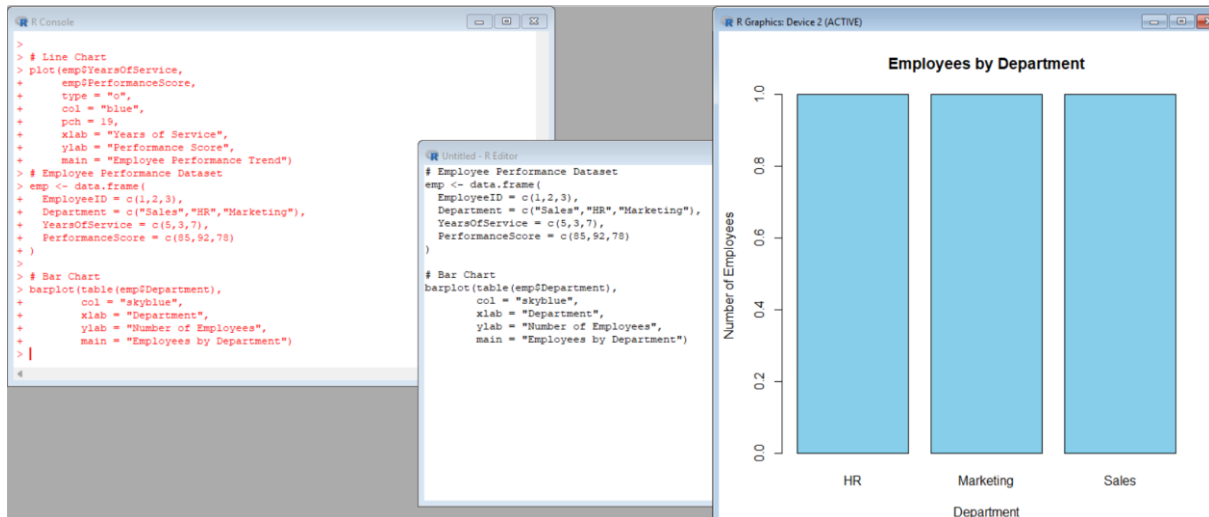
```

```
col = "skyblue",

xlab = "Department",

ylab = "Number of Employees",

main = "Employees by Department")
```



3. Build a table to display the performance data for each employee. Label the table elements.

```
emp <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)

print(emp)
```

	EmployeeID	Department	YearsOfService	PerformanceScore
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

```
library(ggplot2)

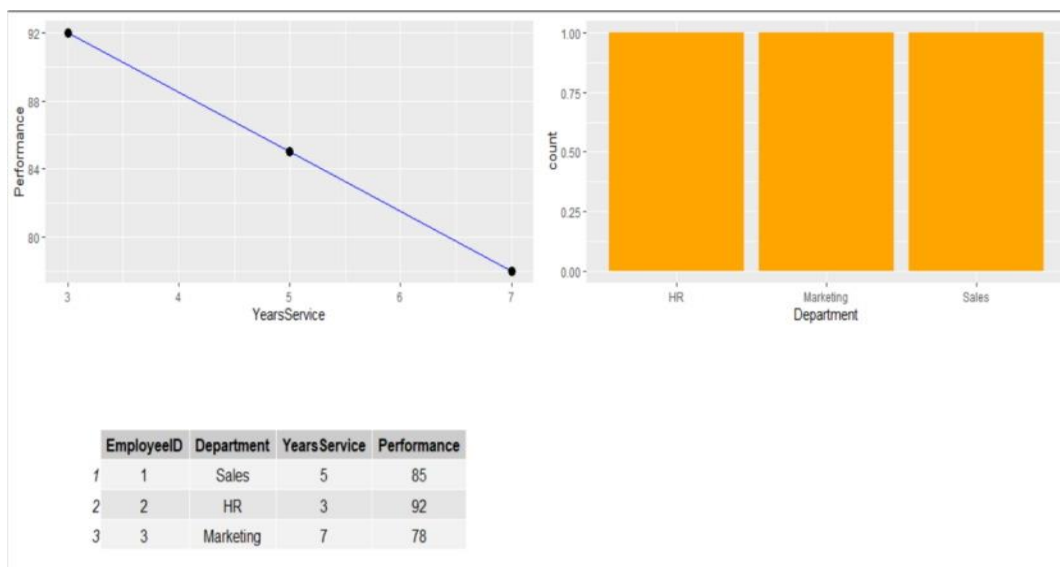
library(gridExtra)

employee <- data.frame(
  EmployeeID=c(1,2,3),
  Department=c("Sales","HR","Marketing"),
```

```

YearsService=c(5,3,7),
Performance=c(85,92,78)
)
p1 <- ggplot(employee,aes(YearsService,Performance,group=1))+
geom_line(color="blue")+
geom_point(size=3)
p2 <- ggplot(employee,aes(Department))+
geom_bar(fill="orange")
table <- tableGrob(employee)
grid.arrange(p1,p2,table,ncol=2)

```



Set 18: Product Inventory Management

Dataset: Product Inventory

Product ID Product Name Quantity Available Price (in \$)

1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```

product <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),

```

```
Quantity = c(250,175,300),
```

```
Price = c(20,15,18)
```

```
)
```

```
barplot(product$Quantity,
```

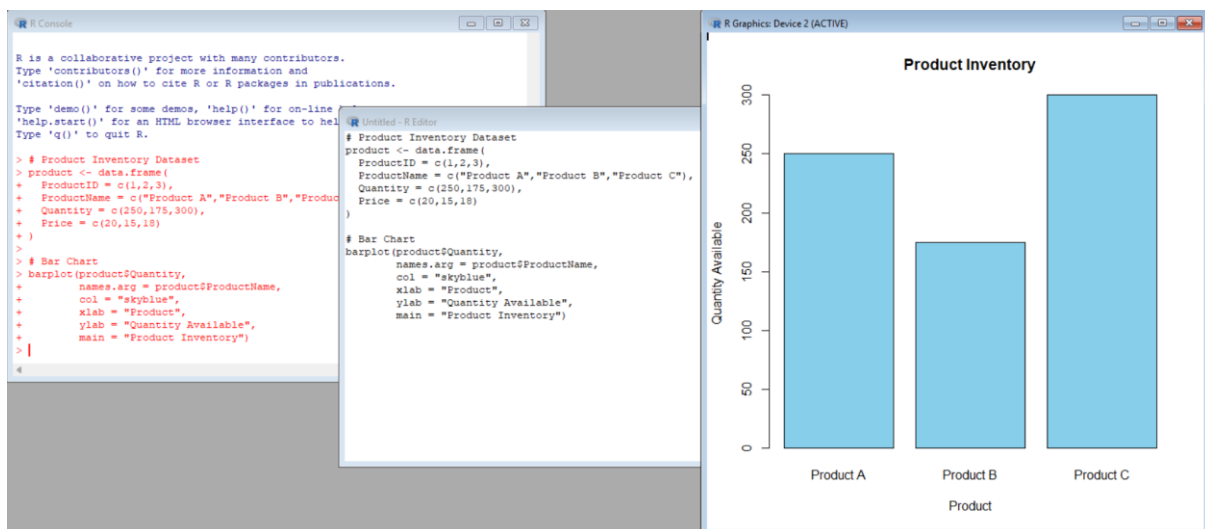
```
names.arg = product$ProductName,
```

```
col = "skyblue",
```

```
xlab = "Product",
```

```
ylab = "Quantity Available",
```

```
main = "Product Inventory")
```



2. Generate a stacked bar chart to show the quantity of each product within different product categories.

```
product <- data.frame(
```

```
ProductName = c("Product A","Product B","Product C"),
```

```
Electronics = c(250,0,0),
```

```
Furniture = c(0,175,0),
```

```
Grocery = c(0,0,300)
```

```
)
```

```
inventory <- rbind(product$Electronics,
```

```
product$Furniture,
```

```
product$Grocery)
```

```
barplot(inventory,
```

```
names.arg = product$ProductName,
```



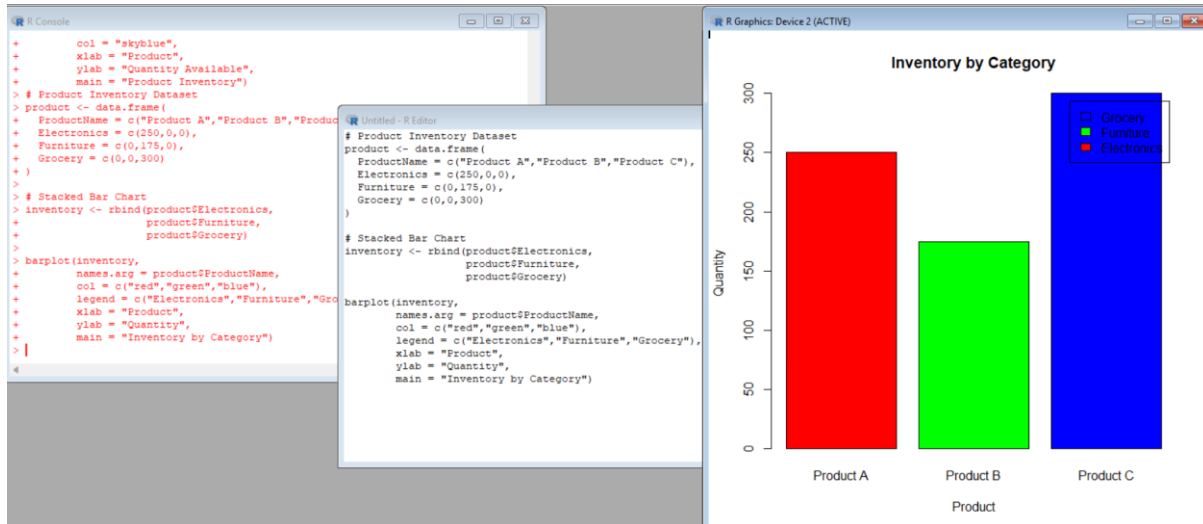
```
col = c("red","green","blue"),

legend = c("Electronics","Furniture","Grocery"),

xlab = "Product",

ylab = "Quantity",

main = "Inventory by Category")
```



3. Build a table to show the inventory data for each product. Label the table elements.

```
product <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  Quantity = c(250,175,300),
  Price = c(20,15,18)
)
print(product)
```

```
ProductID ProductName Quantity Price
1      1    Product A      250     20
2      2    Product B      175     15
3      3    Product C      300     18
```

4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data. library(ggplot2)

```
library(gridExtra)
```

```
product <- data.frame(
  ProductID=c(1,2,3),
  ProductName=c("Product A","Product B","Product C"),
```

```

Quantity=c(250,175,300),
Price=c(20,15,18),
Category=c("Category 1","Category 1","Category 2")
)

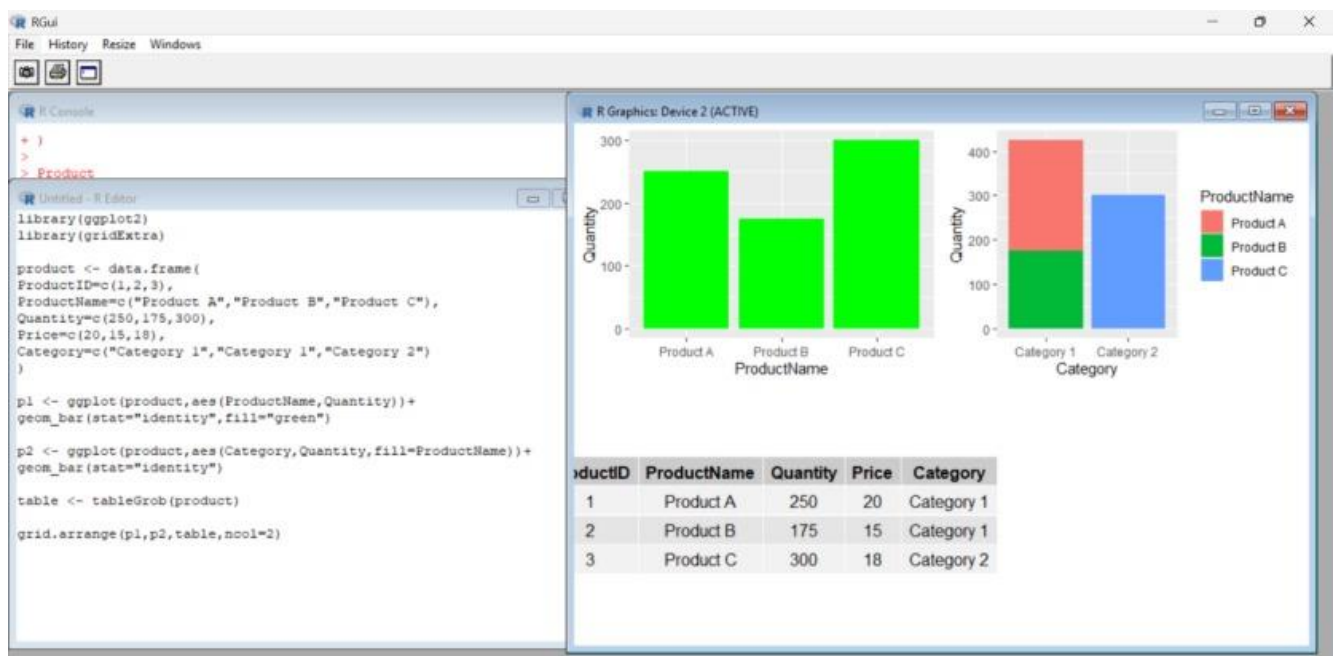
p1 <- ggplot(product,aes(ProductName,Quantity))+
geom_bar(stat="identity",fill="green")

p2 <- ggplot(product,aes(Category,Quantity,fill=ProductName))+
geom_bar(stat="identity")

table <- tableGrob(product)

grid.arrange(p1,p2,table,ncol=2)

```



Set 19: Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

```
library(ggplot2)

survey <- data.frame(

  SurveyID = c(1,2,3),

  Q1 = c("A","B","C"),

  Q2 = c("B","A","A"),

  Q3 = c("C","D","B")

)

ggplot(survey, aes(x = Q1, fill = factor(SurveyID))) +

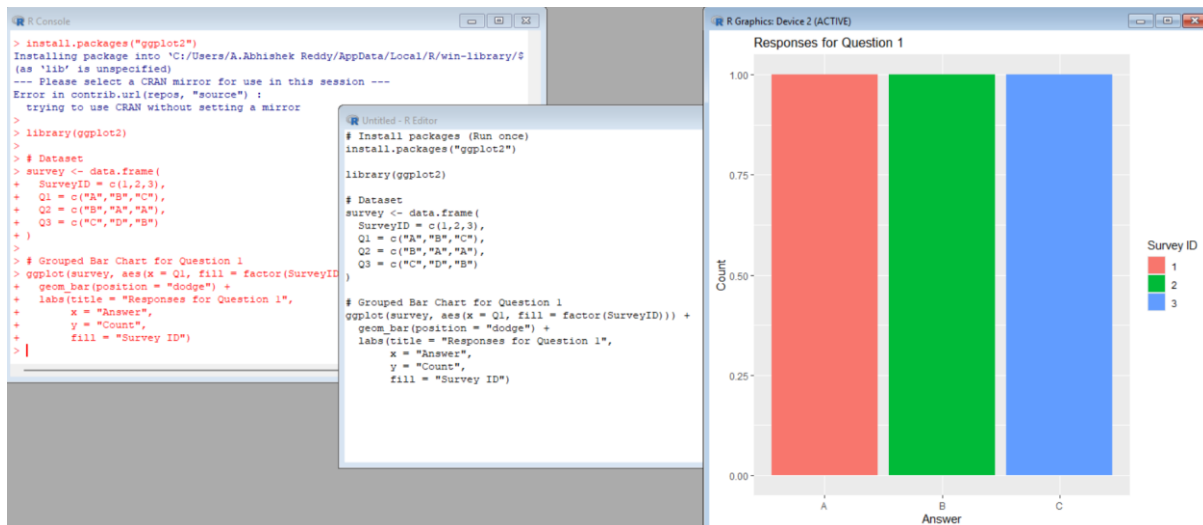
  geom_bar(position = "dodge") +

  labs(title = "Responses for Question 1",

        x = "Answer",

        y = "Count",

        fill = "Survey ID")
```



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
library(ggplot2)

survey <- data.frame(

  SurveyID = c(1,2,3),

  Q1 = c("A","B","C"),

  Q2 = c("B","A","A"),

  Q3 = c("C","D","B")

)
```

```

data <- data.frame(

  Question = rep(c("Q1","Q2","Q3"), each = 3),

  Response = c(survey$Q1, survey$Q2, survey$Q3)

)

ggplot(data, aes(x = Question, fill = Response)) +

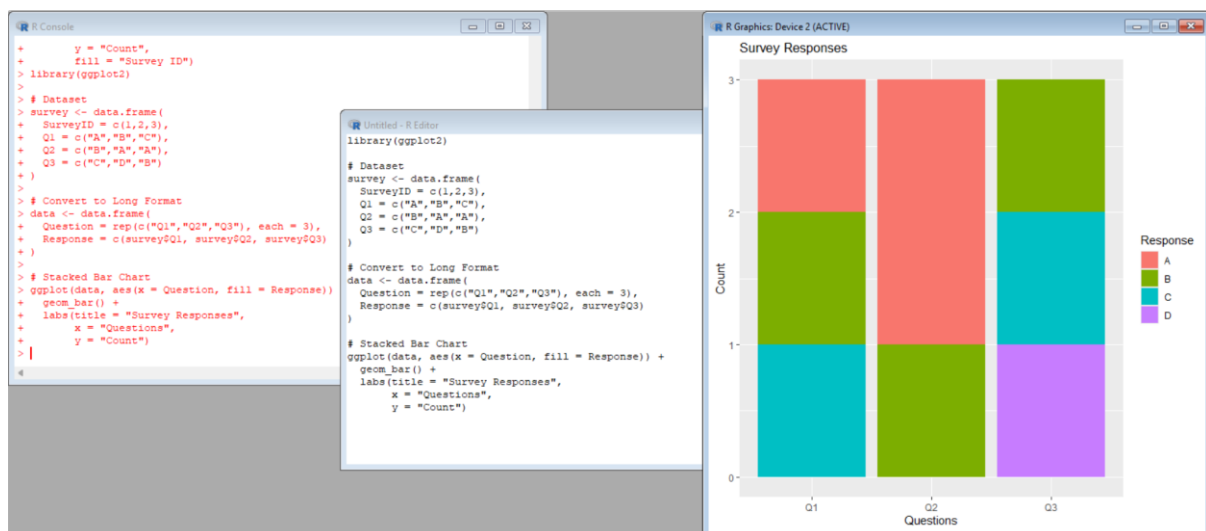
  geom_bar() +

  labs(title = "Survey Responses",

        x = "Questions",

        y = "Count")

```



3. Build a table to show the survey response data for each survey. Label the table elements.

```

survey <- data.frame(

  SurveyID = c(1,2,3),

  Q1 = c("A","B","C"),

  Q2 = c("B","A","A"),

  Q3 = c("C","D","B")

)

```

```
print(survey)
```

```

  SurveyID Q1 Q2 Q3
1         1  A  B  C
2         2  B  A  D
3         3  C  A  B

```

4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

```

library(ggplot2)
library(gridExtra)
library(reshape2)

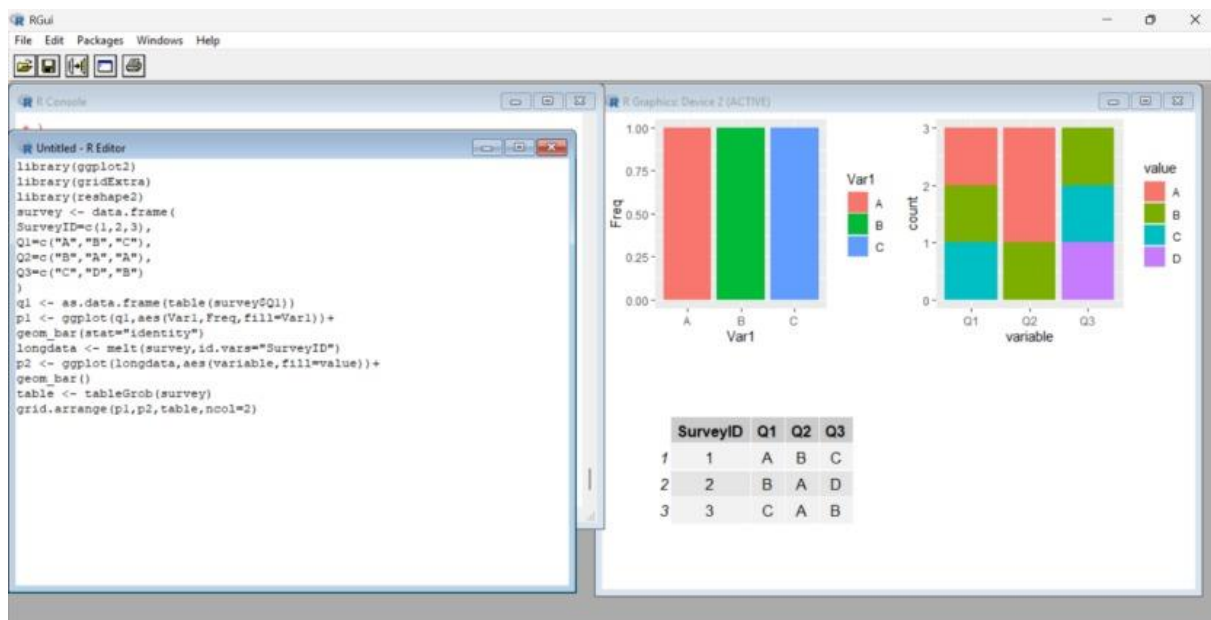
survey <- data.frame(
  SurveyID=c(1,2,3),
  Q1=c("A","B","C"),
  Q2=c("B","A","A"),
  Q3=c("C","D","B")
)

q1 <- as.data.frame(table(survey$Q1))
p1 <- ggplot(q1,aes(Var1,Freq,fill=Var1))+
  geom_bar(stat="identity")

longdata <- melt(survey,id.vars="SurveyID")
p2 <- ggplot(longdata,aes(variable,fill=value))+
  geom_bar()

table <- tableGrob(survey)
grid.arrange(p1,p2,table,ncol=2)

```



Set 20: Stock Analysis

Dataset: Stock Prices

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

```
library(ggplot2)
```

```
stock <- data.frame(
```

```
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
```

```
  StockA = c(100,105,110),
```

```
  StockB = c(150,152,148),
```

```
  StockC = c(120,118,122)
```

```
)
```

```
ggplot(stock, aes(x = Date)) +
```

```
  geom_line(aes(y = StockA, color = "Stock A"), linewidth = 1) +
```

```
  geom_line(aes(y = StockB, color = "Stock B"), linewidth = 1) +
```

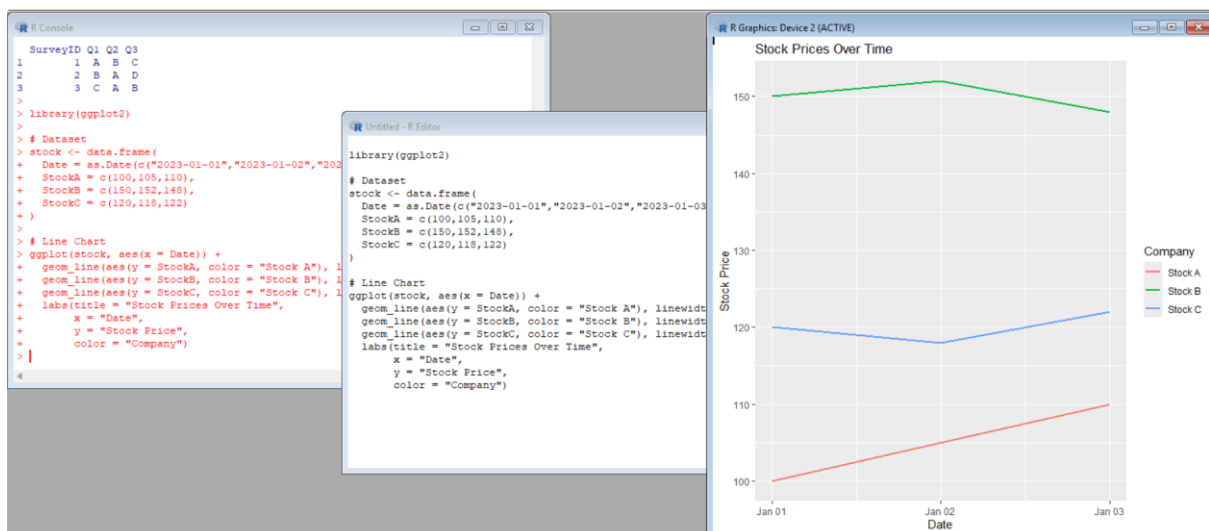
```
  geom_line(aes(y = StockC, color = "Stock C"), linewidth = 1) +
```

```
  labs(title = "Stock Prices Over Time",
```

```
    x = "Date",
```

```
    y = "Stock Price",
```

```
    color = "Company")
```



2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

```
library(ggplot2)

stock <- data.frame(

  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),

  StockA = c(100, 105, 110)

)

change <- c(0, diff(stock$StockA)/head(stock$StockA, -1)*100)

stock$Change <- round(change, 2)

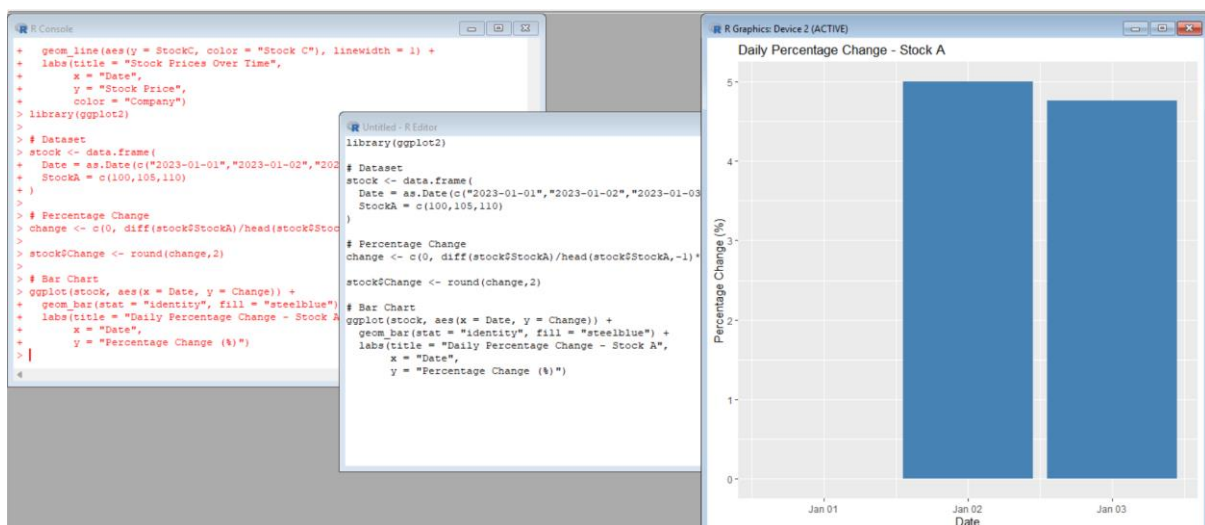
ggplot(stock, aes(x = Date, y = Change)) +

  geom_bar(stat = "identity", fill = "steelblue") +

  labs(title = "Daily Percentage Change - Stock A",

        x = "Date",

        y = "Percentage Change (%)")
```



3. Build a table to display the stock price data for each company over the given period. Label the table elements.

```
stock <- data.frame(

  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),

  StockA = c(100, 105, 110),

  StockB = c(150, 152, 148),

  StockC = c(120, 118, 122)

)
```

```
print(stock)
```

```
      Date StockA StockB StockC
1 2023-01-01    100    150    120
2 2023-01-02    105    152    118
3 2023-01-03    110    148    122
```

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

```
library(ggplot2)
```

```
library(gridExtra)
```

```
library(reshape2)
```

```
stock <- data.frame(
```

```
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
```

```
  StockA=c(100,105,110),
```

```
  StockB=c(150,152,148),
```

```
  StockC=c(120,118,122)
```

```
)
```

```
stock.long <- melt(stock,id.vars="Date")
```

```
p1 <- ggplot(stock.long,aes(Date,value,color=variable))+
```

```
  geom_line()+
```

```
  geom_point()
```

```
change <- c(0,diff(stock$StockA)/head(stock$StockA,-1)*100)
```

```
data <- data.frame(Date=stock$Date,Change=change)
```

```
p2 <- ggplot(data,aes(Date,Change))+
```

```
  geom_bar(stat="identity",fill="purple")
```

```
table <- tableGrob(stock)
```

```
grid.arrange(p1,p2,table,ncol=2)
```